

Labour Market Displacement

GLOSSARY 05 - SHEET 4 / 8

Reduction in labour demand for tasks a technology performs, net of reinstatement

PROFESSIONAL DEFINITION

Labour market displacement is the reduction in demand for labour in tasks that a technology has come to perform. Its net employment effect depends on the balance between the displacement effect, which lowers labour demand at given output, and countervailing effects - the productivity effect, capital accumulation, and above all the reinstatement effect, in which new tasks are created in which labour holds comparative advantage (Acemoglu and Restrepo, 2019).

WHO DEFINED IT ORIGINALLY

The displacement-reinstatement decomposition was developed by Daron Acemoglu of MIT and Pascual Restrepo of Boston University in a sequence of papers from 2018, summarised in the Journal of Economic Perspectives in 2019, with empirical estimates for industrial robots in Acemoglu and Restrepo (2020). Frey and Osborne's (2017) occupational exposure estimates represent the earlier occupation-level tradition.

WHY THIS DEFINITION WAS DEVELOPED

Public debate ran between the claim that automation destroys employment and the claim that it has never done so historically. The decomposition replaces both with an accounting identity in which the net outcome depends on measurable components, and identifies reinstatement as the term that has varied across historical episodes and that is influenced by policy and investment choices.

HOW IT IS USED

Estimate displacement at task level rather than occupation level, then ask separately what new tasks the technology creates and whether the firm or the economy is investing in them; a displacement estimate without a reinstatement estimate is half an answer.

WHEN IT IS USED

In workforce transition planning, in social and economic impact assessment of an AI deployment, in policy analysis, and in works council consultation.

BY WHOM IT IS USED

Policy analysts and labour economists, workforce planners and human resources leaders, employee representatives, and executives accountable for the social impact of deployment decisions.

FIGURE 1 - DISPLACEMENT AGAINST REINSTATEMENT

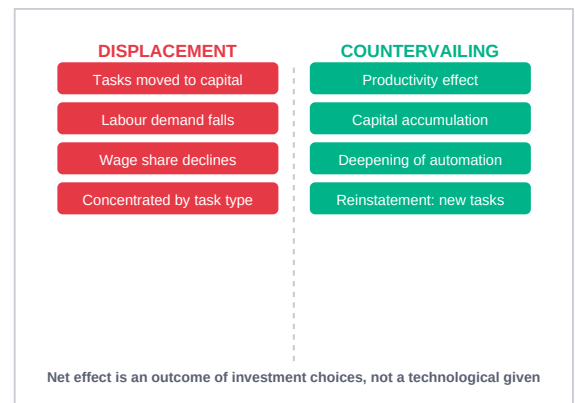


Figure 1. The net employment effect is the balance of the two columns. Historically the right-hand column has dominated, but reinstatement is the term that has varied across episodes and it responds to the direction of investment (after Acemoglu and Restrepo, 2019).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Frey and Osborne's occupation-level probabilities treated whole occupations as automatable and produced headline figures that were widely misread as job-loss forecasts. The task-level decomposition disaggregates the occupation, which is more accurate because most occupations comprise both substitutable and complemented tasks, and it makes reinstatement explicit. The resulting managerial and policy point is that the net effect is partly chosen: technologies can be directed toward automating existing tasks or toward creating new ones.

RELATED DEFINITIONS IN THIS COURSE

Skill-Biased Technological Change - Unit 5

Reskilling and Upskilling - Unit 4

Profit Pool - Unit 3

Externality - Unit 5

Digital Divide - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

- Acemoglu, D. and Restrepo, P. (2019) 'Automation and new tasks: how technology displaces and reinstates labor', Journal of Economic Perspectives, 33(2), pp. 3-30.
- Acemoglu, D. and Restrepo, P. (2020) 'Robots and jobs: evidence from US labor markets', Journal of Political Economy, 128(6), pp. 2188-2244.
- Frey, C.B. and Osborne, M.A. (2017) 'The future of employment: how susceptible are jobs to computerisation?', Technological Forecasting and Social Change, 114, pp. 254-280.